

FINTECH FRAUD MONITORING SYSTEM: REAL-TIME RISK ANALYTICS

1. Project Overview

The "Fintech Fraud Monitoring System" is a comprehensive data engineering and analytics project aimed at identifying and mitigating financial risks in real-time. Traditional fraud detection often overlooks deep feature correlations; this project, however, focuses on **Anomalous Behavioural Patterns** and **Temporal Trends** to secure transaction pipelines. The project follows a rigorous, three-stage analytical pipeline:

- **Phase 1: Exploratory Data Analysis & Cleaning (Jupyter Notebook):** The initial phase involved cleaning and processing a high-volume credit card dataset. Key transformations included mapping raw time offsets into hourly segments and handling PCA-transformed variables to prepare the data for deep analysis.
- **Phase 2: Relational Database Management (PostgreSQL):** To ensure enterprise-grade data integrity, the refined data was migrated to **PostgreSQL (managed via pgAdmin 4)**. This stage allowed for complex SQL querying to filter high-value transactions and validate the dataset before it was connected to the BI layer.
- **Phase 3: Business Intelligence & Visualization (Power BI):** The final stage involved building a high-impact, dark-themed dashboard designed for continuous monitoring. Instead of complex navigation, this project focuses on "Single-View Risk Assessment," where all critical KPIs and correlation charts are visible at once to provide an immediate snapshot of system health. By leveraging Power BI's native **Cross-Filtering** capabilities, users can click on any fraud alert to instantly see its impact across all other visuals—such as the time of attack and specific feature anomalies

Ultimately, this project provides a scalable, data-driven framework that enables financial institutions to move beyond simple rule-based detection and invest in pattern-based fraud prevention.

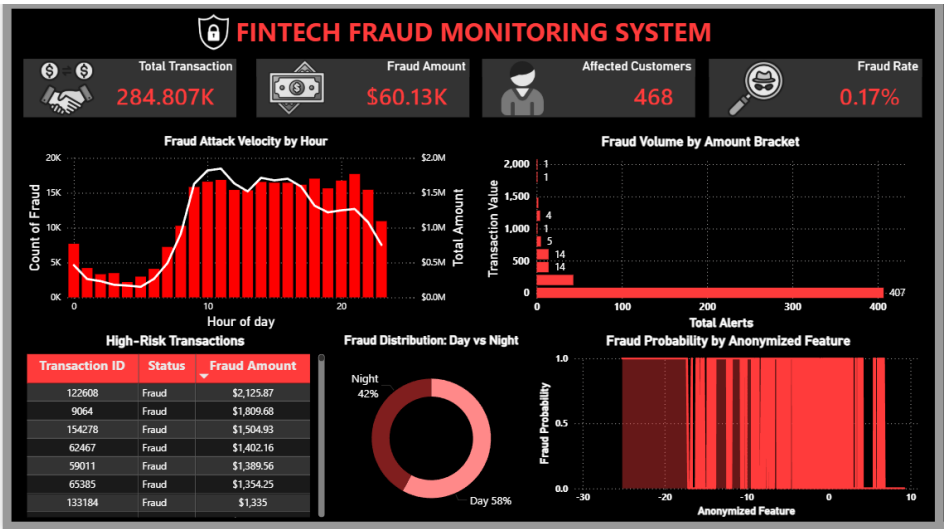


Figure 1: Final Interactive Fraud Monitoring Dashboard Interface

2. Technology Stack

This project utilizes a robust multi-tool architecture to ensure high-performance data processing and interactive visualization:

2.1 Jupyter Notebook (Python 3.x): Used as the primary environment for Exploratory Data Analysis (EDA). Key libraries included **Pandas** for data manipulation, **NumPy** for numerical computations, and **Matplotlib/Seaborn** for initial statistical plotting to identify fraud clusters.

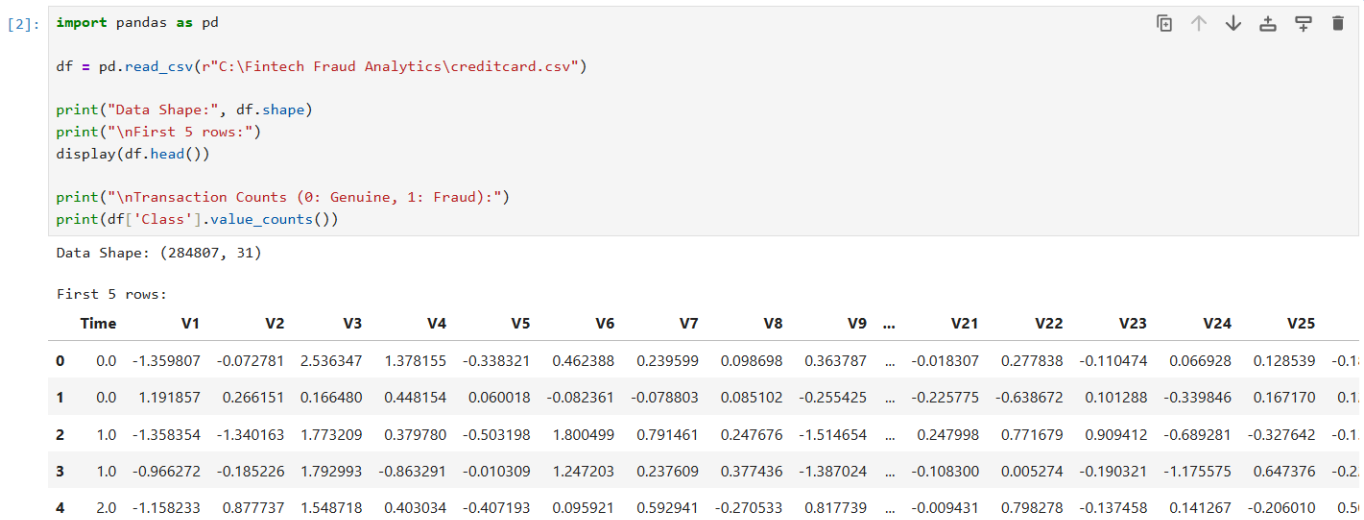


Figure 2: Integrated Technology Stack Jupyter

2.2 PostgreSQL (pgAdmin 4): Employed as the relational database management system (RDBMS) to host the credit card transaction records. SQL was utilized for structured querying, data validation, and ensuring data integrity across large volumes of transaction-level records

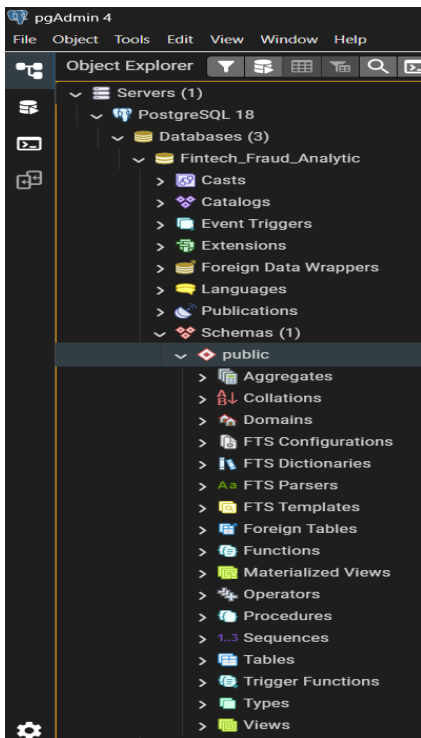


Figure 3: Integrated Technology Stack PostgreSQL

2.3 Power BI Desktop: Used for the final Business Intelligence (BI) layer. This project focuses on high-density data visualization, utilizing Custom DAX Measures for risk probability and Advanced Formatting to create a "Security Command Centre" feel. The integration relies on seamless data connectivity with PostgreSQL to reflect validated transaction records, ensuring that the dashboard acts as a reliable decision-support tool for fraud investigators.

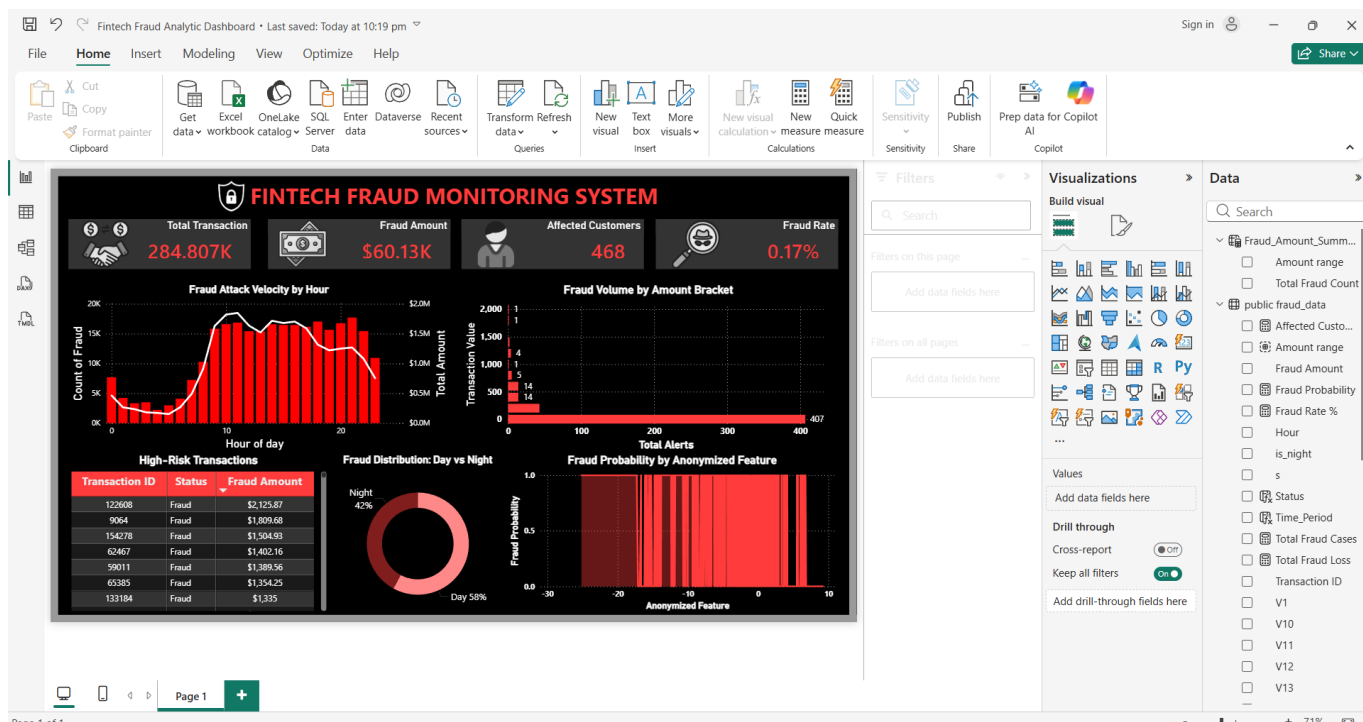


Figure 4: Integrated Technology Stack POWER BI

3. Dataset Summary

he foundation of this analysis is a comprehensive delivery-level dataset of credit card transactions, specifically curated for anomaly detection:

- **Data Source:** Historical Credit Card Transaction Records.
- **Granularity:** Individual transaction-level records, capturing every event (Amount, Time, and Anonymized Features).
- **Key Parameters:**
 - **Anonymized Features (V1-V28):** PCA-transformed components representing sensitive transaction metadata (critical for identifying hidden fraud signatures).
 - **Performance Metrics:** Amount (Transaction Value) and Class (Target variable where 0 = Safe, 1 = Fraud).
 - **Contextual Data:** Time (seconds elapsed since the first transaction in the dataset).
- **Scope:** The dataset comprises thousands of records, providing a statistically significant sample size for scouting and predictive risk analysis.

4. Methodology

The development of the Fraud Monitoring Dashboard followed a systematic four-step analytical workflow:

- **Step 1: Data Acquisition & Pre-processing (Jupyter Notebook):** The raw credit card delivery-level dataset was first loaded into a Jupyter environment. Using the **Pandas** library, initial data cleaning was performed, including the handling of null values and ensuring the integrity of the 'Class' target variable. Feature engineering was conducted to derive metrics such as "**Hourly Transaction Volume**" to align with the temporal analysis objective.

```
print("--- Genuine Transactions Amount Summary ---")
print(df[df['Class'] == 0]['Amount'].describe())

print("\n--- Fraud Transactions Amount Summary ---")
print(df[df['Class'] == 1]['Amount'].describe())

plt.figure(figsize=(10,6))
sns.boxplot(x='Class', y='Amount', data=df)
plt.title('Amount Distribution (Genuine vs Fraud)')
plt.ylim(0, 400)
plt.show()
```

Figure 5: Data Cleaning and Feature Engineering in Python

- **Step 2: Database Schema Design (PostgreSQL):** The processed data was exported as a structured CSV and imported into PostgreSQL via pgAdmin 4. This allowed for a centralized repository where SQL queries were used to validate the data and perform complex aggregations, such as identifying high-value fraud clusters, before connecting to the visualization layer.

```
--Summary of Fraud Transactions

SELECT
    COUNT(*) as total_fraud_cases,
    SUM("Amount") as total_fraud_amount,
    AVG("Amount") as avg_fraud_value
FROM fraud_data
WHERE "Class" = 1;

--Peak Fraud Hours (Time Pattern)

SELECT
    "Hour",
    COUNT(*) as fraud_count
FROM fraud_data
WHERE "Class" = 1
GROUP BY "Hour"
ORDER BY fraud_count DESC
LIMIT 5;
```

Figure 6: Relational Database Management in PostgreSQL

- **Step 3: Business Logic & Metric Development (Power BI - DAX):** After connecting Power BI to the PostgreSQL database, advanced Data Analysis Expressions (DAX) were used to create dynamic measures. Metrics such as Fraud Probability, Total Fraud Amount, and Affected Customer Count were calculated to provide a deeper layer of analysis beyond raw transaction counts.

- **Step 4: UI/UX Optimization (Theme & Layout):** To ensure a professional "Security Command centre" experience, a high-contrast dark theme was implemented. The layout was optimized for Cross-Filtering, allowing users to select a specific time range or amount bracket and instantly see the corresponding impact on feature correlations (V17) and individual high-risk transaction logs.

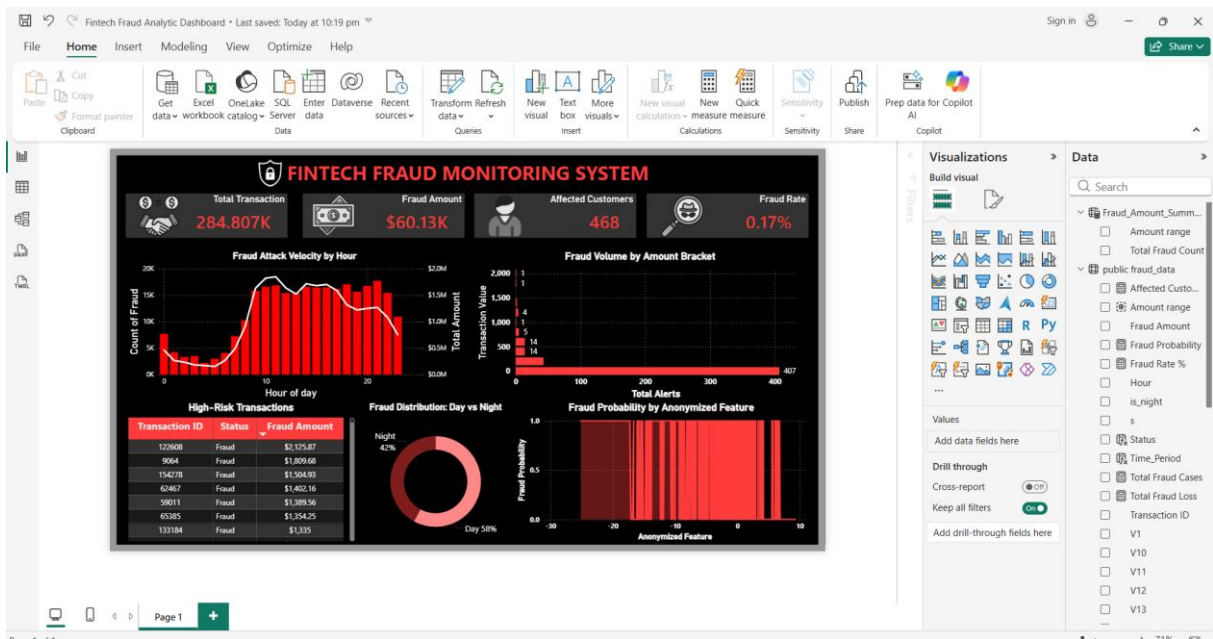


Figure 7: UI/UX Optimization using Cross-Filtering for Deep Dive Analysis

5. Dashboard Visuals & Analysis

This section illustrates the final output of the analytical pipeline. The dashboard is divided into three primary logical zones to provide a 360-degree view of transaction risk:

- **Fraud Attack Velocity (Area/Line Chart):** This visual plot the count of fraud events against the hour of the day. The analysis reveals that fraud attempts are not uniform; they peak during business hours (10 AM - 6 PM), indicating that attackers exploit high transaction volumes to blend in with legitimate users.

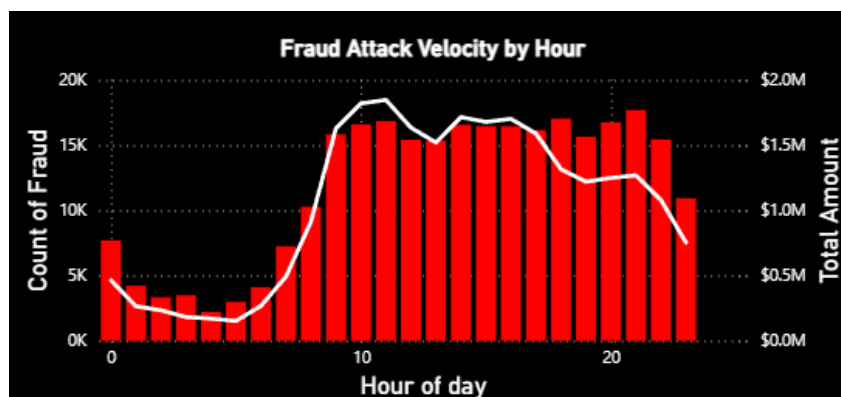


Figure 8: Fraud Attack Velocity

- **Fraud Probability by Anonymized Feature (Area Chart - V17):** A specialized risk curve comparing a key technical feature (V17) and the probability of fraud. The graph clearly shows a

"Danger Zone" where V17 values between -25 and -15 correlate with a 100% fraud probability, serving as a critical threshold for automated blocking systems.

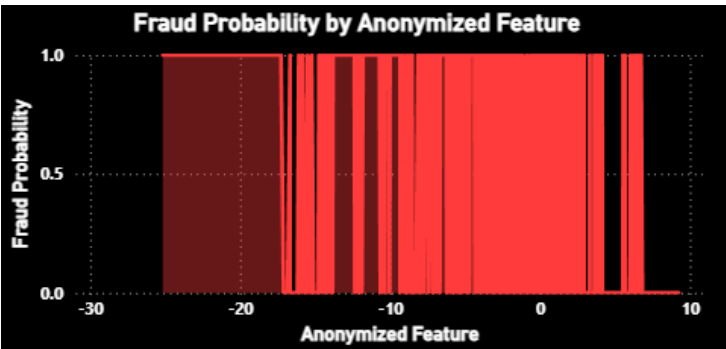


Figure 9: Fraud Probability by Anonymized Feature

- **Fraud Volume by Amount Bracket (Bar Chart):** This bar chart categorizes fraud alerts by transaction value. It helps identify the scammers' strategy—whether they are attempting high-value "Grand Frauds" or multiple low-value "Micro-Frauds" to stay under the bank's radar.

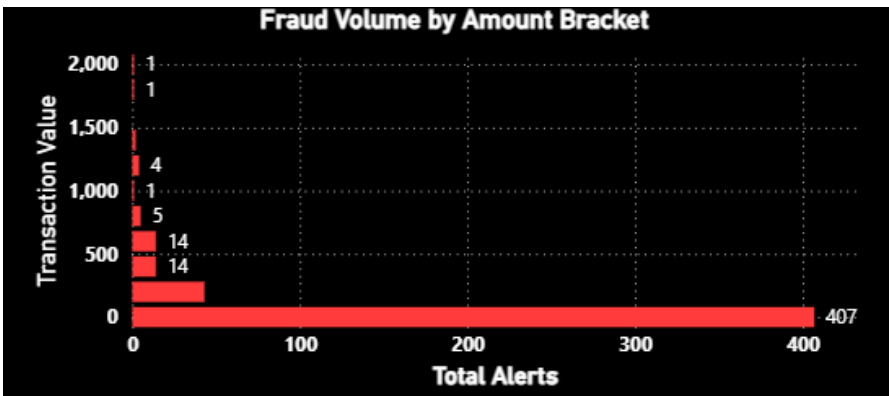


Figure 10: Fraud Volume by Amount Bracket

- **Fraud Distribution: Day vs Night (Donut Chart):** Using a custom calculation, this donut chart visualizes the percentage of attacks occurring in different shifts. With 58% of attacks occurring during the day, it highlights the need for increased monitoring during standard operating hours.

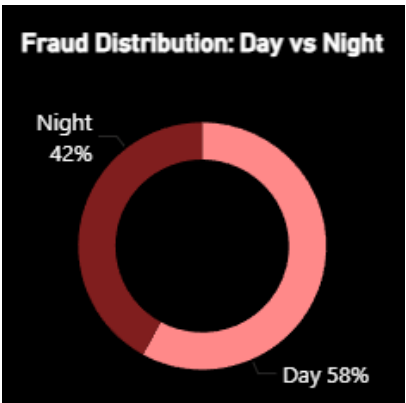


Figure 11: Fraud Distribution: Day vs Night

- **High-Risk Transaction Alerts (Data Table):** This granular list provides real-time details of suspected fraudulent activities, including Transaction IDs and specific Fraud Amounts. By sorting

transactions based on risk severity, it allows investigators to prioritize high-value alerts and take immediate manual or automated action.

High-Risk Transactions		
Transaction ID	Status	Fraud Amount
122608	Fraud	\$2,125.87
9064	Fraud	\$1,809.68
154278	Fraud	\$1,504.93
62467	Fraud	\$1,402.16
59011	Fraud	\$1,389.56
65385	Fraud	\$1,354.25
133184	Fraud	\$1,335

Figure 12: High-Risk Transaction Alerts

6.Business Recommendations

Based on the data-driven insights generated by the dashboard, the following fraud mitigation strategies are recommended:

- 1. **Dynamic Threshold Monitoring:** Implement an automated flagging system that triggers a high-priority alert whenever the V17 feature value drops below -18, as this range has a near-100% correlation with fraudulent activity.
- 2. **Peak-Hour Surveillance:** Increase the frequency of manual audits and system health checks during standard business hours (10:00 AM to 6:00 PM), which have been identified as the primary attack window.
- 3. **High-Value Alert Prioritization:** Use the "High-Risk Transactions Table" to prioritize investigations for transactions exceeding \$2,000, ensuring that large-scale losses are mitigated first.
- 4. **Anomaly-Based Risk Scoring:** Instead of relying solely on transaction amounts, integrate the V17 behavioural pattern into the customer risk score to detect sophisticated "Micro-Frauds" that usually bypass standard filters.

7. Conclusion

The "Fintech Fraud Monitoring System" project successfully demonstrates the power of integrating diverse data technologies—Python, SQL, and Power BI—to solve complex financial security problems. By transitioning from raw, delivery-level data to a highly interactive Business Intelligence tool, we have moved beyond basic rule-based detection.

The project highlights several key achievements:

- **Seamless Data Pipeline:** The successful synchronization between Jupyter Notebook (EDA), PostgreSQL (Data Management), and Power BI (Visualization) ensures that the insights are both accurate and scalable.
- **Deep-Dive Feature Analysis:** The use of specialized risk curves (V17) provides a 360-degree view of fraud signatures, focusing on behavioural anomalies rather than just aggregate stats.
- **Actionable Intelligence:** The dashboard serves as a strategic "Security Command centre," enabling data-backed decisions to protect financial assets and customer trust.